

$$\begin{aligned}
& C_{\phi D}^{(11)} \rightarrow \\
& \frac{1}{16\pi^2} \left(-\frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{p}}] + \frac{1}{12} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,2,-1} [m_d^{\text{r}}, m_q^{\text{p}}] - \frac{1}{6} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \right. \\
& \text{LF}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{p}}] + \frac{1}{12} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1} [m_e^{\text{r}}, m_l^{\text{p}}] + \frac{1}{6} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,1,0} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{3,2,-1} [m_l^{\text{p}}, m_e^{\text{r}}] - \frac{1}{8} \tilde{\mu} y_e^{\text{pr}} \overline{a_e^{\text{pr}}} (3 g_1^2 + g_2^2) \text{LF}_{4,1,-1} [m_l^{\text{p}}, m_e^{\text{r}}] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_e^{\text{pr}} \overline{a_e^{\text{pr}}} \text{LF}_{5,1,-2} [m_l^{\text{p}}, m_e^{\text{r}}] + \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,1,0} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& \frac{1}{6} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_d^{\text{r}}] - \frac{1}{8} \tilde{\mu} y_d^{\text{pr}} \overline{a_d^{\text{pr}}} (5 g_1^2 + 3 g_2^2) \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_d^{\text{r}}] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_d^{\text{pr}} \overline{a_d^{\text{pr}}} \text{LF}_{5,1,-2} [m_q^{\text{p}}, m_d^{\text{r}}] + \frac{1}{12} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,0} [m_q^{\text{p}}, m_u^{\text{r}}] + \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0} [m_q^{\text{p}}, m_u^{\text{r}}] - \frac{1}{24} \tilde{\mu} y_u^{\text{pr}} \overline{a_u^{\text{pr}}} (g_1^2 + 9 g_2^2) \text{LF}_{3,2,-1} [m_q^{\text{p}}, m_u^{\text{r}}] - \\
& \frac{3}{4} \tilde{\mu} g_2^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{4,1,-1} [m_q^{\text{p}}, m_u^{\text{r}}] - \frac{1}{12} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{p}}] - \\
& \frac{1}{12} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{p}}] - \frac{1}{4} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{1}{2} \tilde{\mu} g_1^2 y_u^{\text{pr}} \overline{a_u^{\text{pr}}} \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{p}}] - \frac{1}{8} m_1 \tilde{\mu} g_1^4 \text{LF}_{4,1,-1} [\tilde{\mu}, m_1] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^4 \text{LF}_{5,1,-2} [\tilde{\mu}, m_1] - \frac{3}{8} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{4,1,-1} [\tilde{\mu}, m_2] + \\
& \frac{1}{2} m_2 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{5,1,-2} [\tilde{\mu}, m_2] - \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 (7 m_1 + 5 m_2) \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_1] + \\
& \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \text{LF}_{3,2,1,-2} [m_2, \tilde{\mu}, m_1] + \frac{3}{4} \tilde{\mu} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0} [m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}] + \frac{3}{4} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \\
& \text{LF}_{3,1,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}] + \frac{3}{8} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1} [m_q^{\text{p}}, m_u^{\text{r}}, m_q^{\text{s}}] - \\
& \frac{3}{4} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,1,1,0} [m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}] + \frac{3}{4} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \\
& \text{LF}_{3,1,1,-1} [m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}] + \frac{3}{8} \tilde{\mu} \overline{y_u^{\text{st}}} y_u^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,-1} [m_q^{\text{s}}, m_u^{\text{r}}, m_q^{\text{p}}] + \\
& \frac{1}{2} m_1 \tilde{\mu} g_1^2 g_2^2 \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_2] - \frac{1}{8} \tilde{\mu} g_1^2 g_2^2 (5 m_1 + m_2) \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_2] + \\
& \frac{1}{4} \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \text{LF}_{4,1,1,-2} [\tilde{\mu}, m_1, m_2] + \tilde{\mu} g_1^2 g_2^2 (m_1 + m_2) \text{LF}_{3,2,1,-2} [\tilde{\mu}, m_2, m_1] - \\
& \frac{1}{4} \tilde{\mu}^3 \overline{y_e^{\text{st}}} y_e^{\text{pt}} y_e^{\text{sr}} \overline{a_e^{\text{pr}}} \text{LF}_{2,2,1,1,-1} [m_l^{\text{p}}, m_l^{\text{s}}, m_e^{\text{r}}, m_e^{\text{t}}] - \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}, m_d^{\text{t}}] + \\
& \frac{3}{4} \tilde{\mu} y_d^{\text{sr}} \overline{a_d^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{4} \tilde{\mu}^3 \overline{y_d^{\text{st}}} y_d^{\text{pt}} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_d^{\text{t}}, m_u^{\text{r}}] - \\
& \frac{3}{4} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,1,1,1,0} [m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{4} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{3,1,1,1,-1} [m_q^{\text{p}}, m_q^{\text{s}}, m_u^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{8} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{p}}, m_u^{\text{r}}, m_q^{\text{s}}, m_u^{\text{t}}] + \\
& \frac{3}{8} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{p}}, m_u^{\text{t}}, m_q^{\text{s}}, m_u^{\text{r}}] - \\
& \frac{3}{4} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,1,1,1,0} [m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{4} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{3,1,1,1,-1} [m_q^{\text{s}}, m_q^{\text{p}}, m_u^{\text{r}}, m_u^{\text{t}}] + \\
& \frac{3}{8} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{s}}, m_u^{\text{r}}, m_q^{\text{p}}, m_u^{\text{t}}] + \\
& \left. \frac{3}{8} \tilde{\mu} y_u^{\text{sr}} \overline{a_u^{\text{pr}}} \overline{a_u^{\text{st}}} a_u^{\text{pt}} \text{LF}_{2,2,1,1,-1} [m_q^{\text{s}}, m_u^{\text{t}}, m_q^{\text{p}}, m_u^{\text{r}}] \right)
\end{aligned}$$